

Experiment-4

Aim

To analyze the performance of a **Logistic Regression model** using a numerical dataset by calculating model weights, predictions, accuracy, and confusion matrix using Python.

Algorithm

1. Choose a numerical binary dataset.
 2. Preprocess data (add bias column).
 3. Initialize weights to zero.
 4. Apply **sigmoid** function to convert linear output into probabilities.
 5. Train using **Gradient Descent**:
 - Compute predictions
 - Update weights using gradients
 6. Predict class labels ($\geq 0.5 \rightarrow 1$, else 0)
 7. Compute:
 - Accuracy
 - Confusion matrix (TP, TN, FP, FN)
 8. Display performance results.
-

Input

X (study hours):

[1, 2, 3, 4, 5, 6]

Y (pass/fail):

[0, 0, 0, 1, 1, 1]

✓ Output

```
main.py  [Copy] [Refresh] [Share] [Run] [Clear]

1 import numpy as np
2
3 # -----
4 # 1. CREATE NUMERICAL DATASET
5 # -----
6 np.random.seed(0)
7 x = np.linspace(0, 10, 20)
8 y = 3 * x + 5 + np.random.randn(20) # y = 3x + 5 + noise
9
10 # Train-test split
11 x_train, y_train = x[:15], y[:15]
12 x_test, y_test = x[15:], y[15:]
13
14
15 # -----
16 # 2. LINEAR REGRESSION (MANUAL FORMULA)
17 # -----
18 # Slope (a) and Intercept (b)
19 a = np.sum((x_train - np.mean(x_train)) * (y_train - np.mean(y_train))) / np.sum((x_train - np.mean(x_train))**2)
20 b = np.mean(y_train) - a * np.mean(x_train)
21
22 print("Linear Regression Equation: y =", round(a, 3), "* x", round(b, 3))

# Output
# -----
print("==== Logistic Regression Analysis =====")
print("Weights (Bias, Coefficient):")
print(weights.flatten())

print("\nPredictions:", Y_pred.flatten())
print("Actual:", Y.flatten())

print("\nAccuracy:", accuracy)

print("\nConfusion Matrix:")
print(f"TP = {TP}, FP = {FP}")
print(f"FN = {FN}, TN = {TN}")

=== PERFORMANCE ON TESTING DATA ===
MSE : 0.547601503972162
MAE : 0.5823074780298363
RMSE: 0.740001016196709
R² Score: 0.8358387315025595

=== Code Execution Successful ===
```

✓ Result

The logistic regression model successfully separates the data into two classes with **100% accuracy** for the chosen dataset.

The confusion matrix shows perfect classification with no misclassifications.